

Illuvium Staking V3 Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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October 21, 2025

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Illuvium on Illuvium Staking V3. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated October 21, 2025, was delivered to Illuvium, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 6 days, the Cyfrin team conducted an audit on the Illuvium Staking V3 code provided by Illuvium. The findings consist of 7 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Illuvium Staking V3
Repository	staking-contracts-v3
Commit	c78653ed5f2e...
Fix Commit	f69a4a277ff4...
Audit Timeline	Sep 2nd - Sep 9th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	7	6	1
Informational	8	8	0
Gas Optimizations	1	1	0
Total	16	15	1

6 Audit Scope

The audit scope was limited to:

```
src/L2RevenueDistributorV3.sol
src/StakingVault.sol
src/StakingVaultFactory.sol
```

7 Risk Classification

	Impact: High	Impact: Medium	Impact: Low
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Illuvium Staking V3 provides a revenue distribution and staking system for ILV on Base. Inbound ETH/WETH is converted to ILV via Aerodrome and the ILV is distributed across configured pools. Distribution happens in two phases per call:

1. fixed-BPS pools receive their share
2. the remainder is allocated to reserve-weighted pools based on "units" that reflect either an ILV vault's `StakingVault::totalStaked` or an LP vault's proportional share of ILV reserves in the paired pool

Pools can be vaults (ILV is transferred and `StakingVault::notifyRewardAmount` is called) or directly claimable (ILV accrues to `L2RevenueDistributorV3::pendingIlv` and is later claimed). Staking is handled by time-locked vaults with a per-share accumulator that tracks and pays ILV rewards pro-rata to stake.

8.1 Key Components

- **L2RevenueDistributorV3:** Holds ILV and allocates it across pools. Supports:
  - **Swaps:** `swapExactETHForILV` and `swapExactWETHForILV` via the Aerodrome router
  - **Vault pools:** ILV is transferred to the vault, then `notifyRewardAmount(amount)` is called
  - **Claimable pools:** Increases `pendingIlv`, which is pulled via `claimPool`
  - **Allocation logic:** Fixed-BPS first, then reserve-weighted on the remainder
- **StakingVault:** Time-locked staking for an ERC-20 (ILV or LP) with ILV rewards:
  - Users create per-stake locks; withdrawals require the lock to be expired
  - Rewards are tracked via `accIlvPerShare`; `notifyRewardAmount` increments the accumulator pro-rata to `totalStaked`
  - Users harvest with `claimRewards` (pulls accumulated ILV)
- **StakingVaultFactory:** Deploys new `StakingVault` ERC1967 proxies pointing to a shared implementation and initializes them with the asset, ILV token and lock bounds.

9 Attestation

Cyfrin attests that it performed a security review of the Illuvium Staking V3 code provided by Illuvium.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-10-21-cyfrin-illuvium-stakingv3-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Illuvium has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=f2bf1e175e43875dc870805702c93c68a61e1bdea02d9c6c133b0b6284c348a1
F=2025-10-21-cyfrin-illuvium-stakingv3-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.